

Note 2.2 Properties of Determinants

The Effects of Row Operations on the Determinant

- **Lemma 2.2.1:**

Let A be an $n \times n$ matrix. If A_{jk} denotes the cofactor of a_{jk} for $k = 1, \dots, n$, then

$$a_{i1}A_{j1} + a_{i2}A_{j2} + \dots + a_{in}A_{jn} = \begin{cases} \det(A), & \text{if } i = j \\ 0, & \text{if } i \neq j \end{cases}$$

Proof:

If $i = j$, then the expression is just the [cofactor expansion](#) of $\det(A)$ along the i th row of A . If $i \neq j$, then we can take a matrix B by replacing the j th row of A by the i th row of A :

$$B = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i1} & a_{i2} & \dots & a_{in} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i1} & a_{i2} & \dots & a_{in} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}$$

Since two rows of B are the same, then $\det(B) = 0$. Then

$$\det(B) = a_{i1}B_{j1} + a_{i2}B_{j2} + \dots + a_{in}B_{jn} = a_{i1}A_{j1} + a_{i2}A_{j2} + \dots + a_{in}A_{jn} = 0$$

- **Row Operation 1:**

In general, if A is an $n \times n$ matrix and E_{ij} is the $n \times n$ elementary matrix to do the row operation 1, then

$$\det(E_{ij}A) = -\det(A)$$

- **Row Operation 2:**

If A is an $n \times n$ matrix and $E_i(k)$ is the $n \times n$ elementary matrix to do the row operation 2, then

$$\det(E_i(k)A) = k \det(A) = \det(E_i(k)) \det(A)$$

- **Row Operation 3:**

If A is an $n \times n$ matrix and $E_{ij}(k)$ is the $n \times n$ elementary matrix to do the row operation 3, then

$$\det(E_{ij}(k)A) = \det(A) = \det(E_{ij}(k)) \det(A)$$

- **Summary:**

If E is an elementary matrix, then

$$\det(EA) = \det(E) \det(A)$$

where

$$\det(E) = \begin{cases} -1 & \text{if } E \text{ is of type 1.} \\ k \neq 0 & \text{if } E \text{ is of type 2.} \\ 1 & \text{if } E \text{ is of type 3.} \end{cases}$$

Similar results hold for column operations. The effects that row or column operations have on the value of the determinant can be summarized as follows:

- Interchanging two rows(or columns) of a matrix changes the sign of the determinant.
- Multiplying a single row or column of a matrix by a scalar multiplies the value of the determinant by the scalar.
- Adding a multiple of one row (or column) to another does not change the value of determinant.

Main Results

- **Theorem 2.2.2:**

An $n \times n$ matrix A is singular if and only if

$$\det(A) = 0$$

Proof:

The matrix can be reduced to row echelon form with a finite number of row operations.

Thus,

$$U = E_k E_{k-1} \cdots E_1 A$$

where U is in row echelon form and E_i 's are all elementary matrices. Then

$$\det(U) = \det(E_k \cdots E_1 A) = \det(E_k) \det(E_{k-1}) \cdots \det(E_1) \det(A)$$

since the determinants of the E_i 's are all nonzero, it follows that $\det(U) = 0$ if and only if $\det(A) = 0$. If A is singular, then U has a row consisting entirely of zeros, hence $\det(U) = 0$. If A is nonsingular, then U is upper triangular with 1's along the diagonal and hence $\det(U) = 1$.

Corollary:

If A is nonsingular, it is simpler to reduce A to triangular form. Then we can just evaluate the determinant by the product of the diagonal entries.

Example: Evaluate

$$\begin{vmatrix} 2 & 1 & 3 \\ 4 & 2 & 1 \\ 6 & -3 & 4 \end{vmatrix}$$

Sol:

$$\begin{vmatrix} 2 & 1 & 3 \\ 4 & 2 & 1 \\ 6 & -3 & 4 \end{vmatrix} = \begin{vmatrix} 2 & 1 & 3 \\ 0 & 0 & -5 \\ 0 & -6 & -5 \end{vmatrix} = (-1) \begin{vmatrix} 2 & 1 & 3 \\ 0 & -6 & -5 \\ 0 & 0 & -5 \end{vmatrix} \\ = (-1)(2)(-6)(-5) \\ = -60$$

- **Theorem 2.2.3:**

If A and B are $n \times n$ matrices, then

$$\det(AB) = \det(A) \det(B)$$

Proof:

If B is singular, then AB is also singular, therefore,

$$\det(AB) = 0 = \det(A) \det(B)$$

If B is nonsingular, it can be written as a product of elementary matrices. Then

$$\begin{aligned} \det(AB) &= \det(AE_k E_{k-1} \cdots E_1) \\ &= \det(A) \det(E_k) \cdots \det(E_1) \\ &= \det(A) \det(E_k \cdots E_1) \\ &= \det(A) \det(B) \end{aligned}$$

Lemma: If B is singular, then AB is also singular.

Proof:

If B is singular, then there exists $B\mathbf{x}_0 = \mathbf{0}$, $\mathbf{x}_0 \neq \mathbf{0}$. Thus

$$AB\mathbf{x}_0 = A\mathbf{0} = \mathbf{0}$$

thus there exists a nonzero solution for $AB\mathbf{x}_0 = \mathbf{0}$. Then by the [Theorem 1.5.2](#), AB is singular.

Section 2.2 Exercise

- [Question 5th](#)

Property:

$$\det(\alpha A) = \alpha^n \det(A)$$

- [Question 6th](#)

Property:

$$\det(A^{-1}) = \frac{1}{\det(A)}$$

- [Question 16th](#)

Since $A^T = -A$, $\det(A^T) = \det(A) = \det(-A) = \det(-I) \det(A) = (-1)^n \det(A)$. When n is odd, $\det(A) = -\det(A)$. Then $\det(A) = 0$.

- [Question 18th](#)

- Expanding $\det(E)$ by cofactors along the first row and similarly expanding the others
 $\det(E) = 1 \cdot \det(E_{11}) = 1 \cdot 1 \cdot \det((E_{11})_{11}) = 1 \cdots 1 \det(B) = \det(B)$
- Similar to the first small question but along the last row.
 $\det(F) = 1 \cdot 1 \cdots 1 \cdot \det(A) = \det(A)$.
- Since $C = EF$, $\det(C) = \det(E) \det(F) = \det(B) \det(A) = \det(A) \det(B)$.

- [Question 19th](#)

Exchange the each corresponding rows for A and B , which totally exchange k times. That is

$$M = E_k E_{k-1} \cdots E_1 C$$

where $C = \begin{pmatrix} A & O \\ O & B \end{pmatrix}$. Thus $\det(M) = \det(E_k) \cdot \det(E_{k-1}) \cdots \det(C) = (-1)^k \det(A) \det(B)$.